Detecting AI-Generated Text Using Lightweight Linguistic Models

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Introduction

The rapid rise of large language models (LLMs) such as ChatGPT has made it increasingly difficult to distinguish between human-written and AI-generated text.

This project explores whether **simple, interpretable models** using linguistic features can reliably detect AI-generated content.

Research Questions:

- Can lightweight models distinguish AI vs human text?
- What linguistic patterns differentiate the two?
- How well does performance generalize across sources?

Methodology

Dataset:

- HC3 dataset (Human vs ChatGPT responses)
- Multiple domains: Reddit, StackExchange, etc.

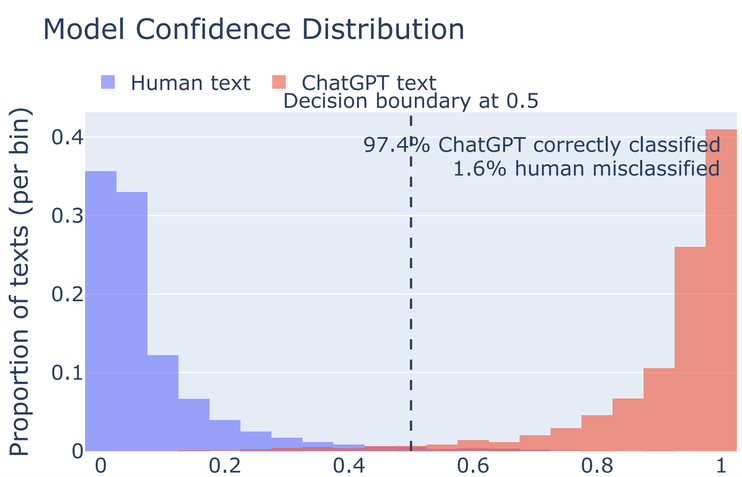
Model:

- TF-IDF vectorization
- Logistic Regression classifier
- Interpretable linear weights

Evaluation:

- Accuracy
- ROC-AUC
- Source-wise generalization

Model Confidence Distribution



Key Insight

Even a simple linear model can achieve **strong separation between human and AI-generated text**, suggesting that LLM outputs exhibit consistent and detectable linguistic patterns.

Accuracy by Source

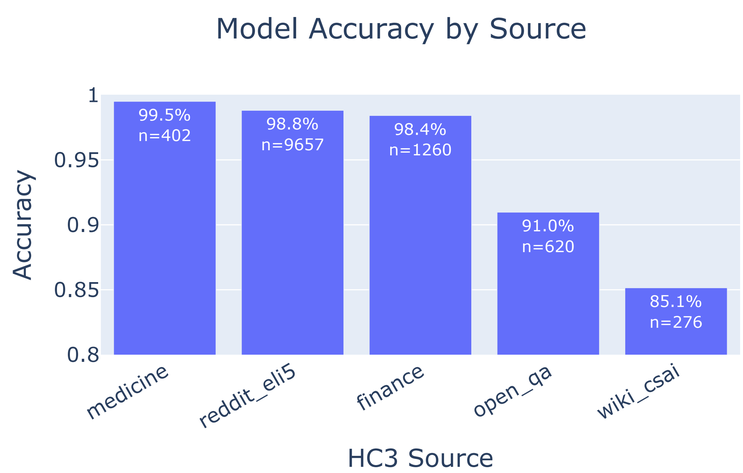


Figure 2. Model accuracy across different data sources. Performance remains consistently high, indicating strong generalization.

ROC Curve

Feature Importance

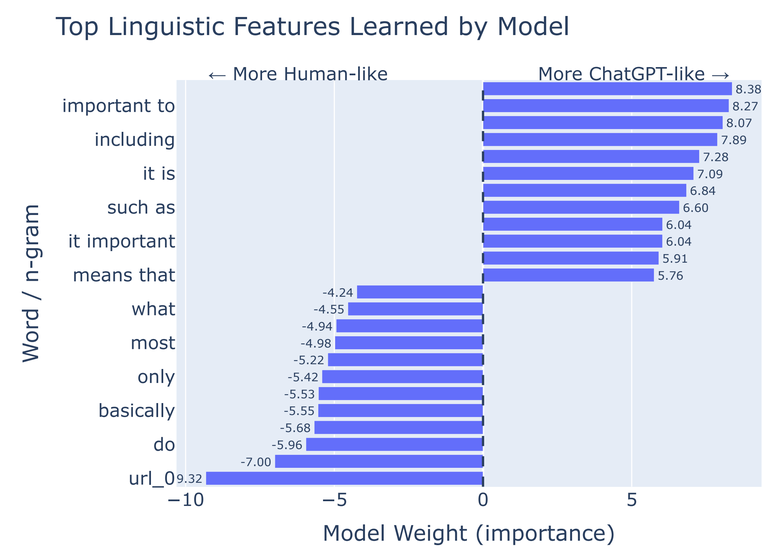


Figure 4. Top linguistic features learned by the model. Positive weights indicate ChatGPT-like patterns; negative weights indicate human-like patterns.

Findings

Key Observations:

- AI-generated text is highly consistent in style
- Certain phrases and structures strongly signal LLM output
- Human text exhibits greater variability

Implications:

- Lightweight models remain competitive for detection
- Interpretability is a major advantage over deep models
- Detection may become harder as LLMs improve

Future Work

- Evaluate robustness on newer LLMs (GPT-4+, Claude, etc.)
- Explore adversarial paraphrasing attacks
- Incorporate syntactic and semantic features
- Compare against deep neural classifiers

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